# Analysis Übungsstunde 5

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## Komplexe Fourier Reihe

normale Fourier-Reihe: $a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right) + b_n \sin\left(\frac{n\pi}{L}x\right)$

ersetze durch cplx schreibweise $\sin(x) = \frac{1}{2i}\left(e^{ix} - e^{-ix}\right)$

$\cos(x) = \frac{1}{2}\left(e^{ix} + e^{-ix}\right)$

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \tfrac{1}{2}\left(e^{i\frac{n\pi}{L}x} + e^{-i\frac{n\pi}{L}x}\right) + b_n \tfrac{1}{2i}\left(e^{i\frac{n\pi}{L}x} - e^{-i\frac{n\pi}{L}x}\right)$$

$\boxed{\frac{1}{i} = -i}$ $\left(\frac{1}{i}\cdot\frac{i}{i} = \frac{i}{i^2} = \frac{i}{-1}\right)$

$$= a_0 + \sum_{n=1}^{\infty} \underbrace{\frac{a_n - ib_n}{2}}_{c_n} e^{i\frac{n\pi}{L}x} + \underbrace{\frac{a_n + ib_n}{2}}_{c_{-n}} e^{-i\frac{n\pi}{L}x}$$

$$= a_0 + \sum_{n=1}^{\infty} c_n e^{i\frac{n\pi}{L}x} + c_{-n} e^{-i\frac{n\pi}{L}x}$$

↙ passe einfach auf dass $c_n$ $\forall n \in \mathbb{Z}$ geht
bsp: $\frac{1}{n}$ → $c_0$ separat

$$= c_0 + \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} c_n \cdot e^{i\frac{n\pi}{L}x} \quad \text{oder} \quad \sum_{n=-\infty}^{\infty} c_n \cdot e^{i\frac{n\pi}{L}x}$$

(I) ⟺ (II)

$$f(x) = c_0 + \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} c_n \cdot e^{\frac{i\pi n}{L}x}$$

❗ Wie immer, $f(x)$ muss 2L-Periodisch sein!

$$c_n = \frac{1}{2L}\int_{-L}^{L} f(x)\cdot e^{-\frac{i\pi n}{L}x} \quad ; \quad c_0 = \frac{1}{2L}\int_{-L}^{L} f(x)dx$$

$$a_0 = c_0 \; ; \; a_n = c_n + c_{-n} \; ; \; b_n = i(c_n - c_{-n})$$

$$e^{ix} + e^{-ix} = 2\cos(x); \qquad e^{ix} - e^{-ix} = 2i\sin(x)$$

$c_n = \frac{1}{2}(a_n - ib_n)$

$c_{-n} = \frac{1}{2}(a_n + ib_n)$

Remember:

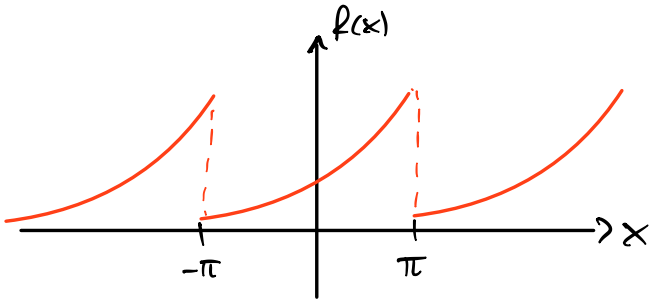
- $e^{\pm ix} = \cos(x) + i\sin(x)$
- $e^{i\pi} = -1$ ($\cos(\pi) + i\sin(\pi)$, $\cos(\pi) = -1$, $\sin(\pi) = 0$) → $e^{\pm in\pi} = (-1)^n \quad n \in \mathbb{Z}$

# Bsp: Kplx Fourier-Reihe

Berechne Fourier Reihe von $f(x) = e^x$, $x \in [-\pi, \pi]$

mit Periode $f(x) = f(x + 2\pi)$

$p = 2\pi \doteq 2L \qquad L = \pi$




$$f(x) = c_0 + \sum_{\substack{n=-\infty \\ n\neq 0}}^{\infty} c_n \cdot e^{\frac{in\pi}{L}x}$$

$$\Rightarrow c_0 = \frac{1}{2L}\int_{-L}^{L} f(x)\,dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^x\,dx = \frac{1}{2\pi}\left[e^{\pi} - e^{-\pi}\right] = \frac{\sinh(\pi)}{\pi}$$

$$\Rightarrow c_n = \frac{1}{2L}\int_{-L}^{L} f(x)\cdot e^{-i\frac{n\pi}{L}x}\,dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^x \cdot e^{-inx}\,dx \qquad \left(\tfrac{1}{2}(e^x - e^{-x}) \hat{=} \sinh(x)\right)$$

$$= \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{(1-in)x}\,dx = \frac{1}{2\pi}\left[\frac{1}{(1-in)}\cdot e^{(1-in)x}\right]_{-\pi}^{\pi}$$

$$= \frac{1}{2\pi}\cdot\frac{1}{(1-in)}\left(e^{\pi - in\pi} - e^{-\pi + in\pi}\right) = \frac{1}{2\pi}\cdot\frac{1}{(1-in)}\left(e^{\pi}\cdot e^{-in\pi} - e^{\pi}\cdot e^{+in\pi}\right)$$

$$= \frac{1}{2\pi}\cdot\frac{1}{(1-in)}\left(e^{\pi}\cdot(-1)^n - e^{-\pi}(-1)^n\right) = \frac{(-1)^n}{2\pi(1-in)}\left(e^{\pi} - e^{-\pi}\right)\cdot\frac{2}{2}$$

$$= \frac{(-1)^n}{\pi}\cdot\sinh(\pi)\cdot\underbrace{\frac{1}{(1-in)}\cdot\frac{(1+in)}{(1+in)}}_{=(1+n^2)} = c_n$$

$$f(x) = \overbrace{\frac{\sinh(\pi)}{\pi}}^{c_0} + \sum_{\substack{n=-\infty \\ n\neq 0}}^{n=\infty} \overbrace{\frac{(-1)^n}{1+n^2}\cdot\frac{\sinh(\pi)(1+in)}{\pi}}^{c_n} \cdot e^{inx} \quad (I)$$

$f(x) = \sum c_n \cdot e^{i\frac{n\pi}{L}x}$ nicht vergessen

oder $$\frac{\sinh(\pi)}{\pi}\sum_{n=-\infty}^{n=\infty}(-1)^n\frac{(1+in)}{1+n^2}\cdot e^{inx} \quad (II)$$

# Bsp. Prüfung Sommer 2017 A2

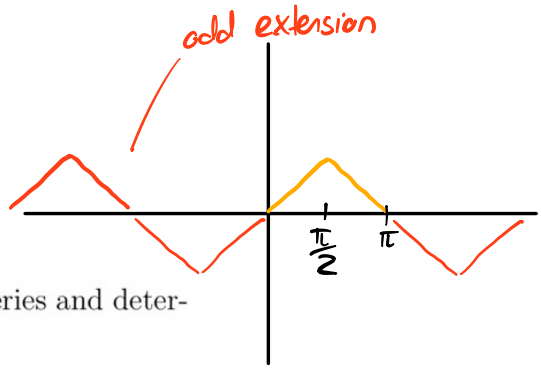
**2. Fourier Series** *(6+1=7 Points)*

a) Given the odd, $2\pi$-periodic function $f : \mathbb{R} \to \mathbb{R}$ with

$$f(x) = \begin{cases} x & 0 \le x \le \frac{\pi}{2} \\ -(x-\pi) & \frac{\pi}{2} \le x \le \pi, \end{cases}$$

determine the real Fourier series of $f$, i.e. state it's Fourier series and determine the coefficients.

b) Does the Fourier series converge pointwise to the function $f$? Please justify your answer.




→ odd : $a_0 = 0$ , $a_n = 0$ ✓

→ $2\pi$ periodic $\quad 2L = 2\pi \rightarrow L = \pi$

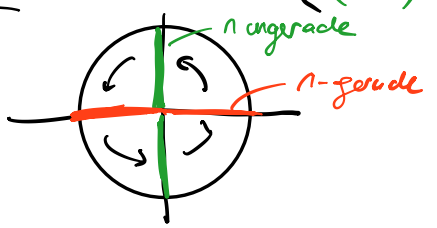
$$b_n = \frac{2}{L}\int_0^L f(x)\cdot \sin\left(\frac{n\pi}{L}x\right)dx \qquad L=\pi$$

$$= \frac{2}{\pi}\left( \int_0^{\frac{\pi}{2}} x\cdot\sin(nx)\,dx + \int_{\frac{\pi}{2}}^{\pi} -(x-\pi)\cdot\sin(nx) \right)$$

$$\int x\sin(nx) = -\frac{x}{n}\cos(nx) + \frac{1}{n}\int\cos(nx)dx$$
$$= -\frac{x}{n}\cos(nx) + \frac{1}{n^2}\sin(nx) = \left[\frac{\sin(nx) - nx\cos(nx)}{n^2}\right]$$ ← Grenzen einsetzen

$$= \frac{4}{\pi n^2}\cdot\sin\left(n\frac{\pi}{2}\right) \qquad \sin\left(n\frac{\pi}{2}\right) = \begin{cases} 0 & n=2k \\ (-1)^k & n=2k+1 \end{cases}$$

$$= \frac{4}{\pi(2k+1)^2}\cdot(-1)^k$$




$$f(x) = \sum_{n=1}^{\infty} b_n \sin(nx) = \sum_{k=0}^{\infty} \frac{4\cdot(-1)^k}{\pi(2k+1)^2}\cdot\sin((2k+1)x)$$

→ $n=1 = 2k+1 \rightarrow k=0$

FALLS: (hypothetisch)

$$= \begin{cases} \frac{1}{k} & n = 2k \\ (-1)^k & n = 2k+1 \end{cases} \qquad L = \pi$$

$$f(x) = \sum_{k=1}^{\infty} \frac{4}{\pi(2k)^2} \cdot \frac{1}{k} \cdot \sin(2kx) + \sum_{k=0}^{\infty} \frac{4}{\pi(2k+1)^2} \cdot (-1)^k \cdot \sin((2k+1)x)$$

Alle geraden: $n = 2 = 2k \rightarrow k = 1$

ungeraden Terme: $n = 2k+1 \rightarrow k = 0$

Für Serie: Brauche auch Skript letzte Woche!